PHYS 161 Notes

Miagao

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1 Units, Physical Quantities, and Vectors

1.1 SI Units

- **SI Units** refer to system of units used by engineers and scientists across the world.
- For PHYS 161, these will be the most commonly used units in problems.

Application	Unit	Shorthand
Time	Seconds	s
Length	Meter	m
Mass	Kilogram	kg

- Additionally, many prefixes may be used in problems, necessitating conversions.

Length	Mass	Time
1 nanometer $= 1$ nm $= 10^{-9}$ m	1 microgram $= 1\ \mu$g $= 10^{-9}$ kg	1 nanosecond$= 1$ ns $= 10^{-9}$ s
1 micrometer $= 1\ \mu$m $= 10^{-6}$ m	1 milligram $= 1$ mg $= 10^{-6}$ kg	1 microsecond $= 1\ \mu$s $= 10^{-6}$ s
1 millimeter $= 1$ mm $= 10^{-3}$ m	1 gram $= 1$ g $= 10^{-3}$ kg	1 millisecond $= 1$ ms $= 10^{-3}$ s

1.2 Unit Conversions

- In equations, quantities may only be added if they have the same dimension.
- When multiplying two quantities, the product's dimension is the product of the operands' dimensions.

$$\left(\frac{\mathrm{m}}{\mathrm{s}}\right)(\mathrm{s}) = \mathrm{m}$$

1.3 Significant Figures

- The **uncertainty** of a measurement is often indicated by the number of important digits, known as the **significant figures.**
  - In provided values, all non-zero digits are significant figures.

$$525 \Rightarrow \text{three significant figures}$$

  - Meanwhile, zero digits are only significant if they are between non-zero digits, or they are trailing non-zero digits when there is a decimal point.

$$40001 \Rightarrow \text{five significant figures}$$
$$120.0 \Rightarrow \text{four significant figures}$$

- When adding quantities, the significant figures of the sum is equal to the operand with the least significant figures.

$$25.7 + 13 = 38.7 \approx 39 \Rightarrow \text{two significant figures}$$

- When multiplying quantities, the same rule applies to the product.

$$(13.7)(20.95) = 287.015 \approx 287 \Rightarrow \text{three significant figures}$$

- In longer physics problems with numerous equations, using the actual values in calculations and only converting the final answer to the correct amount of significant figures is best practice.

2 Motion Along a Straight Line

2.1 Displacement, Time, Velocity, and Acceleration

- **Time** is measured in seconds, and usually refers to the time passed between two particular points in time.
  - Denoted t.
- **Displacement** and **distance** is measured in meters.
  - Denoted Δx and x, respectively, along the x-axis.
- **Velocity** is measured in meters per second.
  - Denoted v.
- **Acceleration** is measured in meters per square second.
  - Denoted a.

$$\Delta x = x_2 - x_1$$
$$v_{\text{av}} = \frac{\Delta x}{\Delta t}$$
$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}$$
$$a_{\text{av}} = \frac{\Delta v}{\Delta t}$$
$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{\mathrm{d}v}{\mathrm{d}t}$$

2.2 Motion with Constant Acceleration

- With constant acceleration, the following **kinematic equations** hold true:

$$v = v_0 + a\Delta t$$
$$\Delta x = v_0 \Delta t + \frac{1}{2} a \Delta t^2$$
$$v^2 = v_0^2 + 2a\Delta x$$

Derivation: Displacement Kinematic Equation

$$v = v_0 + a\Delta t$$
$$\int_{t_1}^{t_2} v \,\mathrm{d}t = \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t$$
$$\int_{t_1}^{t_2} \frac{\mathrm{d}x}{\mathrm{d}t} \,\mathrm{d}t = \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t$$
$$\int_{x_1}^{x_2} \mathrm{d}x = \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t$$
$$\Delta x = v_0 \Delta t + \frac{1}{2} a \Delta t^2$$

3 Motion in Two or Three Dimensions

3.1 Vector Quantities

- Many of the definitions from one-dimensional motion hold true in two or three dimensions, albeit using vectors.

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$
$$\vec{v} = v_x\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k}$$
$$\vec{a} = a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k}$$

3.2 Projectile Motion

- In **projectile motion**, an object in the air only experiences acceleration due to gravity.
- Because the same kinematic equations can be applied *per component* of the vectors, the following generalizations can be made (given that z is the vertical direction):

$$v_x = v_{0x}$$
$$\Delta x = v_{0x}\Delta t$$

$$v_y = v_{0y}$$
$$\Delta y = v_{0y}\Delta t$$

$$v_z = v_{0z} - g\Delta t$$
$$\Delta z = v_{0z}\Delta t - \frac{1}{2}g\Delta t^2$$
$$v_z^2 = v_{0z}^2 - 2g\Delta z$$

Constant g is the acceleration due to Earth's gravity, $9.81\ \mathrm{m/s^2}$. Note that y is generally the vertical direction when dealing with two dimensions.

3.3 Circular Motion

- Even if a particle's speed is constant, its velocity cannot be constant in circular motion because *its direction is constantly changing.*
- The direction of the velocity changes due to acceleration directed toward the center of its rotation, known as **centripetal acceleration.**
  - Denoted a_t or a_{rad}.
  - Thus, when centripetal acceleration is the only acceleration vector acting on a particle, the particle undergoes **uniform circular motion** because it will always be perpendicular to the particle's velocity vector.
- An acceleration vector parallel to the particle's velocity vector is known as **tangential acceleration.**
  - Denoted a_t.
  - This is known as **nonuniform circular motion**, because the speed of the particle will also be changing.

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t}$$

- **Frequency** is the number of revolutions (full rotations) per second.
  - Denoted f.
- **Period** is the number of seconds needed to make one revolution.
  - Denoted T.

$$T = \frac{1}{f}$$

- Given radius of the circular motion R

$$v = \frac{2\pi R}{T}$$
$$a_c = \frac{v^2}{R}$$

3.4 Relative Motion

- Given reference frames A, B, and C where AB means "A in the reference frame of B"

$$\vec{v}_{AB} + \vec{v}_{BC} = \vec{v}_{AC}$$
$$\vec{v}_{AB} = -\vec{v}_{BA}$$

- The **Law of Sines** and the **Law of Cosines** are useful formulas for relative motion problems whenever angles are needed.

Definition: Law of Sines

Given sides a, b, and c opposite of angles A, B, and C, respectively,

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Definition: Law of Cosines

Given sides a, b, and c opposite of angles A, B, and C, respectively,

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = a^2 + c^2 - 2ac\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

4 Newton's Laws of Motion

4.1 Newton's First Law

Definition: Newton's First Law

An object subject to no external forces is at rest or moves with constant velocity if viewed from an inertial reference frame.

$$\text{Inertial Reference Frame} \equiv \text{Reference frame in which Newton's laws are valid}$$

- An inertial reference frame must not be accelerating or rotating.

4.2 Newton's Second Law

- **Forces** are pushes or pulls on objects, measured in Newtons (N) or $(\mathrm{kg} \cdot \mathrm{m/s}^2)$.

Definition: Newton's Second Law

$$\Sigma \vec{F} = m\vec{a}$$

4.3 Newton's Third Law

Definition: Newton's Third Law

For every action there is an equal and opposite reaction.

Given objects A and B where AB means "exerted from A on B"

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

- For example, when two objects collide, they experience the same magnitude of force from the collision.

5 Applying Newton's Laws

5.1 Types of Forces

- **Fundamental forces** are the four natural interactions that govern how particles interact.
  - **Weight**
  - Electromagnetic
  - Weak nuclear
  - Strong nuclear
- **Non-fundamental forces** are forces whose interactions are derived from the fundamental forces.
  - **Spring**
  - **Contact** (tension, normal, friction, ...)
  - Microscopic
- The **weight** force is from gravity. On Earth, it is defined as follows:

$$\vec{w} = m\vec{g}$$

- However, there are also exists a general definition for weight force, the **Universal Law of Gravitation** using the gravitational constant G

$$G = 6.6743 \times 10^{-11}\ \mathrm{m}^3 \cdot \mathrm{kg}^{-1} \cdot \mathrm{s}^{-1}$$

Definition: Universal Law of Gravitation

The force between any two objects is defined as follows (given that r is the distance between the objects' centers):

$$F_1 = F_2 = G\frac{m_1 m_2}{r^2}$$

- The **spring force** is from compressing a spring.
  - Springs have differing **spring constants** usually denoted as k, and they define the ratio between the force exerted by the spring and the distance compressed from the equilibrium position.

$$\vec{F}_{\text{spr}} = -k\vec{x}$$

- The **normal force** is exerted by a surface on an object.
  - Generally denoted $\vec{n}$.
- The **tension force** is the pulling force exerted equally throughout a rope, cable, etc.
  - Generally denoted $\vec{T}$.
- The **friction force** is exerted by a surface on an object and acts perpendicular to the normal force. It depends on the roughness of the surface (expressed by the coefficient of friction).
  - **Kinetic friction**, denoted $\vec{f}_k$ with coefficient of kinetic friction μ_k, acts on objects sliding across a surface.

$$\vec{f}_k = \mu_k \vec{n}$$

  - **Static friction**, denoted $\vec{f}_s$ with coefficient of static friction μ_s, acts on still objects until the applied force exceeds the maximum static friction.

$$\vec{f}_s \leq \vec{f}_{s,\max} = \mu_s \vec{n}$$

5.2 Forces in Equilibrium

- Objects are in equilibrium when they are at rest or moving with constant velocity, meaning that the net force acting on those objects must be zero.

$$\begin{aligned} \Sigma \vec{F} &= 0 \\ &\Rightarrow \\ \Sigma F_x &= 0 \\ \Sigma F_y &= 0 \end{aligned}$$

- Thus, unknown forces can be solved in terms of other forces very easily.

5.3 Dynamics of Circular Motion

- Based on Newton's Second Law, circular motion can result from many different forces.

$$\Sigma F = ma_c = \frac{mv^2}{R}$$

- As a result, denoting a singular "centrifugal force" is generally not accurate.
- Additionally, direction should be considered relative to the center of rotation.

6 Work and Kinetic Energy

6.1 Work

- A force does **work** on a body if the body undergoes a displacement.
- The units of work are joules (J) or newton-meters.

Definition: Work

Given work W, force $\vec{F}$, and displacement $\vec{s}$

$$W = \vec{F} \cdot \vec{s} = Fs\cos(\theta)$$

- Thus, it follows that a force does **negative work** when applied opposite of the direction of $\vec{s}$ and does **zero work** when applied perpendicular to the direction of $\vec{s}$.
- Additionally, given this definition, there are multiple ways to define **net work.**

$$W_{\text{net}} = \Sigma\vec{F} \cdot \vec{s} = \Sigma W$$

6.2 Kinetic Energy

- **Energy** is the capacity to do work, and also uses joules.
- **Kinetic energy** is the energy of a body from its mass and speed.

Definition: Kinetic Energy

Given kinetic energy K, mass m, and speed v

$$K = \frac{1}{2}mv^2$$

- Because kinetic energy is based off of speed, it is either positive or zero.
- Additionally, work may be expressed as the **change in kinetic energy.**

$$W = \Delta K$$

Derivation: Work-Energy Theorem

Given work W and kinetic energy K

$$F_{\parallel} = ma = m\frac{v_2^2 - v_1^2}{2s}$$

$$F_{\parallel}s = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

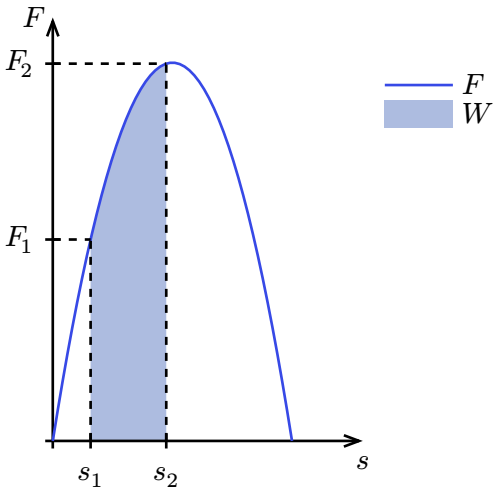
$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \Delta K$$

- However, these formulas only hold true when the force remains constant during the displacement.

6.3 Work and Energy with Varying Force

- If an object is experiencing a varying net force, then the total work can be approximated using the average force per increment of the total displacement.
- Thus, as the increment tends to zero, work may be defined as follows:

$$W = \int_{s_1}^{s_2} F \, \mathrm{d}s$$



6.4 Power

7 Potential Energy and Energy Conservation

Example: Rollercoaster

A rollercoaster starts at a height of